



Indian Institute of Science Education and Research Mohali

CHM203 v.2 - Chemical kinetics

MS-1 Examination, February 7, 2026

Total marks: 20; Time 12.00 noon – 1.00 pm

(Important note: Please write your Name and Roll No. on the Q-paper; please return the Q-Paper along with the answer sheet.)

1. Derive a microscopic expression for the ideal gas law, and prove that $pv = \frac{2}{3} n N_a < \epsilon >$, where N_a is Avogadro number (5 Marks)
2. Bees like honey. A sphere of radius r_0 is coated with honey and hanging in a tree. Bees are attracted to the honey such that the average number of bees per unit volume is given by Kr^{-5} , where K is a constant and r is the distance from the center of the sphere. Derive the normalized distribution function for the bees. They can be at any distance from the honey, but they cannot be inside the sphere. Using this distribution, calculate the average distance of a bee from the center of the sphere. (5 Marks)
[Hint: Find out the normalization constant K by the equation $\int P(Q)dQ = 1$; Please consider the same on a 3D spherical coordinate.]
3. What are the eight possible mechanisms of ester hydrolysis? Represent each mechanism using a schematic diagram. Out of the eight possible mechanisms, how many of them are experimentally observed and studied? (2+2+1 Marks)
4. *p*-nitrophenylglutamate undergoes hydrolysis at 300K. The reaction is monitored spectrophotometrically at 400 nm in a 1 cm path-length cuvette. If the A_α is 0.634 at pH 7.14, calculate the first-order rate constant. (3 Marks + 2 marks for graph)

t(sec)	A_t	t(sec)	A_t
0	0.0	120	0.402
40	0.265	180	0.451
80	0.342	α	0.634

[Hint: Use proper graph paper to estimate the rate constant.]